

Homework 4

Mth 742

2/12/2026

1. For a normal distribution with known variance, i.e. $X_1, X_2, \dots, X_n$ are iid $N(\mu, \sigma^2)$ where σ^2 is known. Use the Cramér–Rao lower bound (CRLB) to obtain a lower bound for the variance of any unbiased estimator of μ .
2. Let $X_1, X_2, \dots, X_n$ be an iid sample from $\text{Bernoulli}(p)$.
 - a) Derive the Fisher information $I(p)$.
 - b) Show that $I(p) \rightarrow \infty$ as $p \rightarrow 0$ or $p \rightarrow 1$.
3. For the normal model in Question 1:
 - a) Compute $\mathbb{V}(\bar{X})$.
 - b) Compare it to the CRLB.
 - c) Conclude whether $\bar{X}$ is efficient.
4. For the Bernoulli model, consider $T = X_1$ as an estimator of p .
 - a) Is T unbiased? Justify your answer.
 - b) Compute $\mathbb{V}(T)$.
 - c) Compare the variance in part b) to the CRLB and comment on efficiency.
5. Let $X_1, X_2, \dots, X_n$ be an iid sample from $\text{Exponential}(\lambda)$. whose pdf is:

$$f(x|\lambda) = \lambda e^{-\lambda x}$$

- a) Determine $I(\lambda)$.
- b) Let $\theta = \log(\lambda)$. Compute $I(\theta)$
- c) Verify the transformation rule,

$$I(\theta) = I(\lambda) \left(\frac{d\lambda}{d\theta} \right)^2$$

6. Give an example of a statistical model where the CRLB exists but no estimator attains it.